PGCP-AC

INTRODUCTION TO APPLIED GENERATIVE AI: FROM LLMS TO AGENTIC AI

Foundations of AI, NLP & Modern LLMs

Dr. Kiran Waghmare
CDAC Mumbai



What is Artificial Intelligence?

- **Artificial Intelligence (AI) refers to computer systems capable of performing tasks that normally require human intelligence.**

- **Examples:**
  - Image recognition
  - Speech recognition
  - Language understanding
  - Decision making
  - Content generation



What is Artificial Intelligence?

- **Artificial Intelligence (AI) refers to computer systems capable of performing tasks that normally require human intelligence.**

- **Examples:**
  - Image recognition
  - Speech recognition
  - Language understanding
  - Decision making
  - Content generation

Artificial : man made (created by human)

Intelligence :Learn,understand, Think, solve problems, make decision

Adap from experience



MAJOR TASKS OF ARTIFICIAL INTELLIGENCE

AI enables machines to learn, reason, perceive, communicate and make decisions like humans.

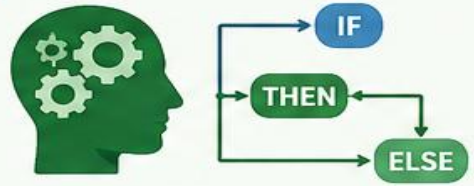
1 LEARNING



Acquiring knowledge from data and experience.

Example: Predicting house prices using historical data.

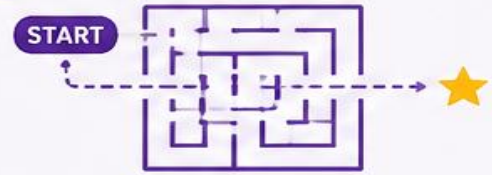
2 REASONING



Making logical decisions and drawing conclusions.

Example: Medical diagnosis systems.

3 PROBLEM SOLVING



Finding solutions to complex problems.

Example: Route optimization in navigation systems.

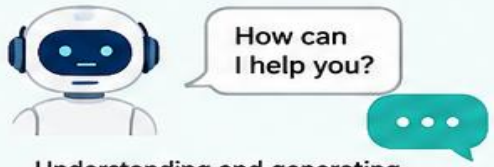
4 KNOWLEDGE REPRESENTATION



Organizing and storing information for decision-making.

Example: Expert systems and knowledge graphs.

5 NATURAL LANGUAGE PROCESSING (NLP)



Understanding and generating human language.

Example: Chatbots, translation systems, virtual assistants.

6 COMPUTER VISION



Interpreting and analyzing images and videos.

Example: Face recognition and object detection.

7 SPEECH RECOGNITION



Converting spoken language into text and understanding speech.

Example: Voice assistants and speech-to-text applications.

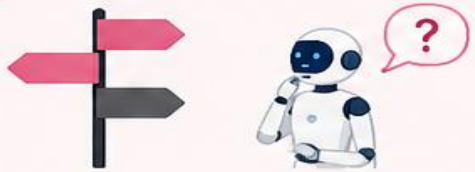
8 PLANNING AND SCHEDULING



Creating action plans to achieve specific goals.

Example: Airline scheduling and logistics management.

9 DECISION MAKING



Selecting the best action among alternatives.

Example: Financial risk assessment systems.

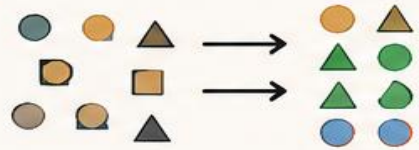
10 ROBOTICS



Controlling physical machines to perform tasks autonomously.

Example: Industrial robots and self-driving vehicles.

11 PATTERN RECOGNITION



Identifying patterns and trends in data.

Example: Fraud detection and handwriting recognition.

12 PREDICTION AND FORECASTING

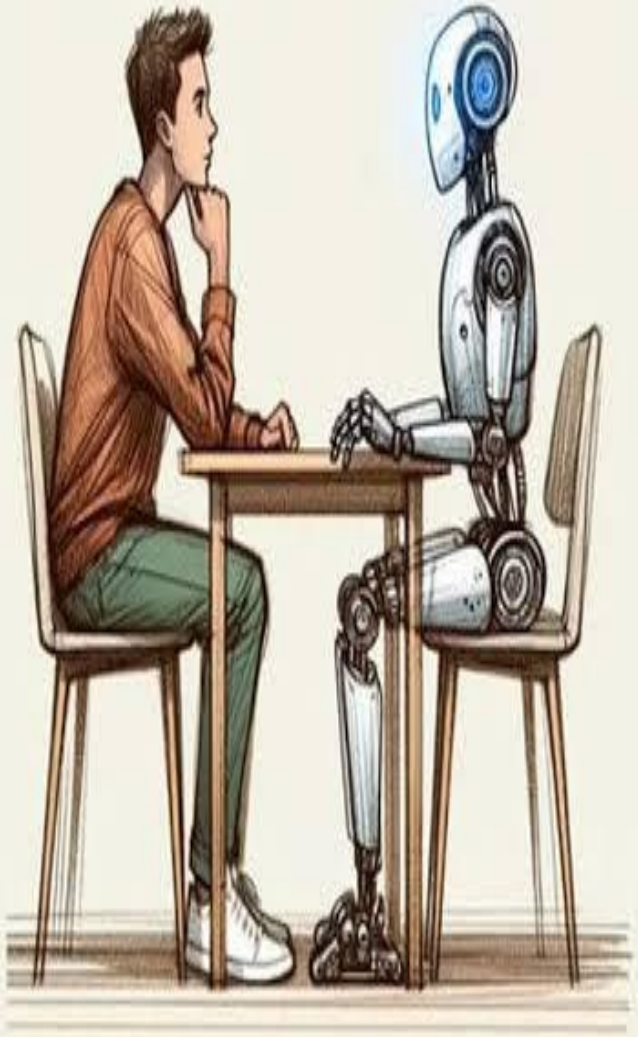


Estimating future events based on historical data.

Example: Weather forecasting and demand prediction.

‘Turing Test’

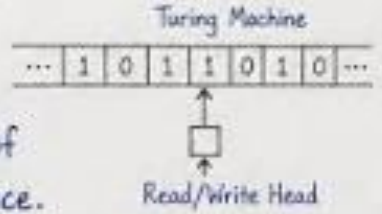


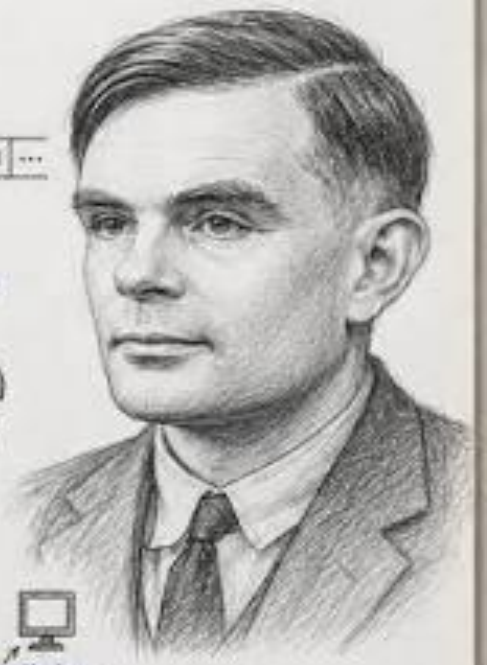


Alan Turing (1912–1954)

Father of Computer Science & AI

- Proposed the Turing Machine concept.

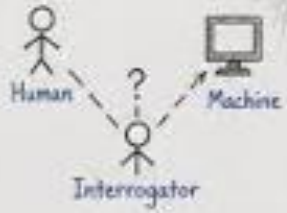




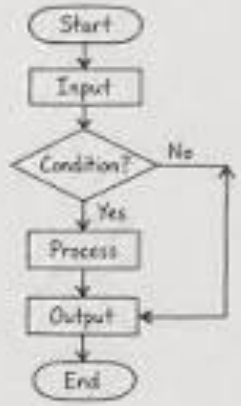
- Laid the foundation of modern computer science.
- Helped break the German Enigma code during World War II.



- Introduced the Turing Test.



- Shaped ideas in artificial intelligence and algorithms.



- His work changed the world of technology.



EDUSPHERE ACADEMY

*D:\KGenAI\GenAI Notes.txt - Notepad++

File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?

new 7 | new 8 | API Keys.txt | new 9 | GenAI Notes.txt

-Autonomy

Judge(Human)

Human Participant

Machine

Adv:
Machine intelligence
AI research
intelligent behaviour

Limitations:
measure imitation
no logical reasons

1950 :Turing Test
1956 : Birth of AI
1980 : Expert System
1997 : Deep Blue : chess program
2006 : Deep Learning : Neurons
2017 : Transformer : NLP
2022 : LLM : ChapGPT : Genrative AI
2024 : Agentic AI :

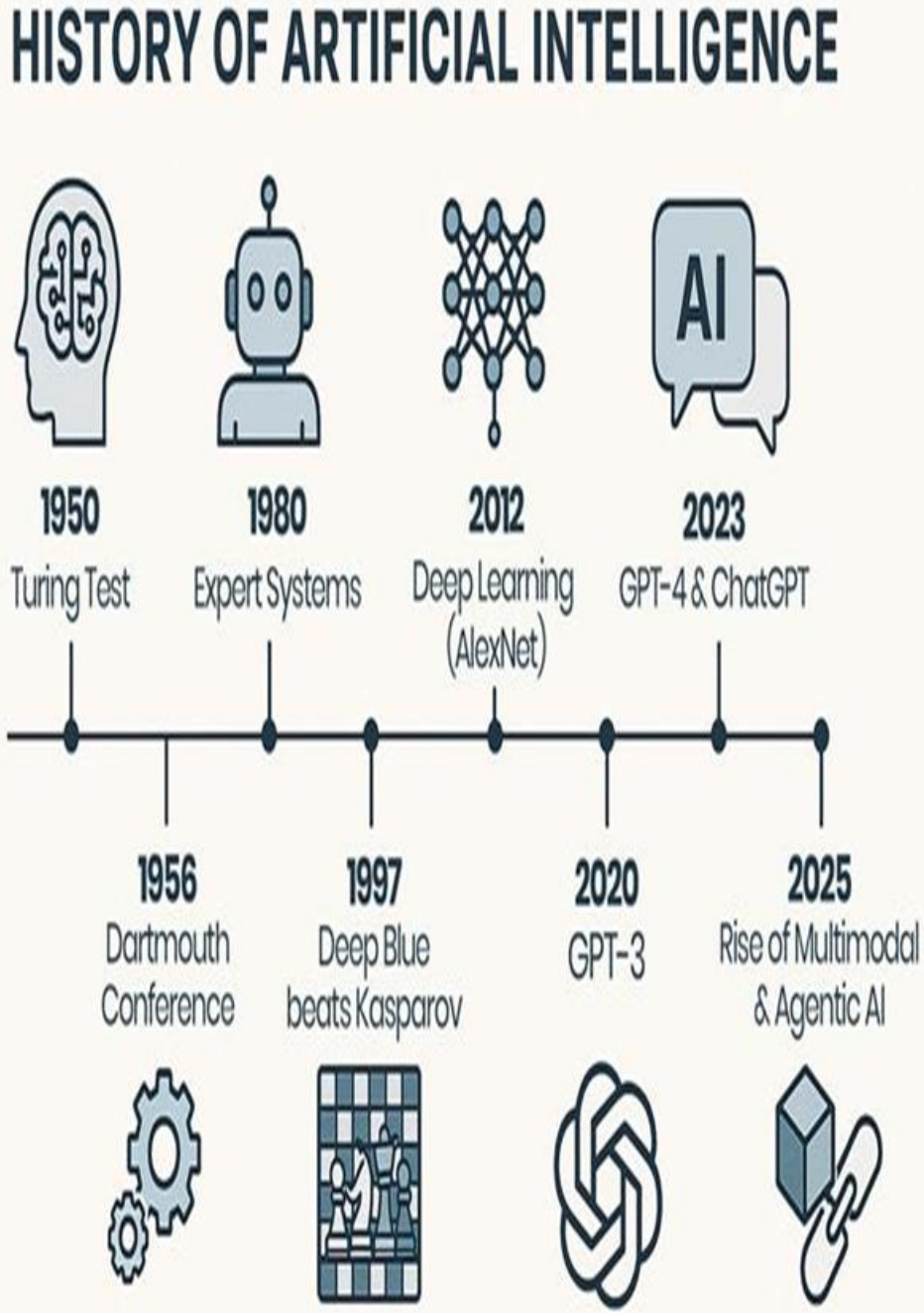
Timeline of AI Evolution

Year	Milestone	Key Contribution
1950	Turing Test	Test for machine intelligence
1956	Birth of AI	AI became an academic discipline
1960–1980	Early AI Research	Symbolic reasoning and problem solving
1980–1995	Expert Systems	Knowledge-based decision making
1997	Deep Blue	AI defeated the world chess champion
2006	Deep Learning	Revival of neural networks
2017	Transformers	Revolutionized Natural Language Processing
2022	ChatGPT	Popularized Generative AI
2024–Present	Agentic AI	AI systems capable of autonomous planning and tool use

AI Evolution Timeline

Era	Technology	Example
1950s	Rule-based AI	Expert Systems
1980s	Machine Learning	Decision Trees
2010s	Deep Learning	CNN, RNN
2017	Transformers	Attention Mechanism
2022+	Generative AI	ChatGPT, Claude





HISTORY OF ARTIFICIAL INTELLIGENCE (AI)

YEAR	MILESTONE	DETAILS	EXAMPLE / IMPACT
1950	Turing Test	British mathematician Alan Turing proposed the Turing Test to determine whether a machine can exhibit intelligent behavior similar to a human.	If a human cannot distinguish between a machine and another human in a conversation, the machine is considered intelligent.
1956	Birth of AI	The term Artificial Intelligence (AI) was officially introduced at the Dartmouth Conference by John McCarthy, marking the beginning of AI as a research field.	Researchers developed early AI programs capable of solving simple mathematical and logical problems.
1980	Expert Systems	AI systems were designed to mimic the decision-making ability of human experts using predefined rules and knowledge bases.	MYCIN, an expert system, helped doctors diagnose bacterial infections and recommend antibiotics.
1997	IBM Deep Blue	IBM's supercomputer Deep Blue defeated world chess champion Garry Kasparov, proving AI could outperform humans in complex strategy games.	Deep Blue became the first computer to beat a reigning world chess champion in a tournament.
2012	Deep Learning Revolution	Deep learning gained popularity due to advances in neural networks, large datasets, and powerful GPUs.	AlexNet won the ImageNet competition with record-breaking image recognition accuracy, transforming computer vision.
2017	Transformers	Google researchers introduced the Transformer architecture in the paper *Attention Is All You Need*. It enabled faster and more accurate language understanding.	Transformers became the foundation for modern AI models such as BERT, GPT, and many large language models (LLMs).
2022	ChatGPT	OpenAI launched ChatGPT, bringing conversational AI to millions of users. It can answer questions, generate text, write code, summarize information, and assist with learning.	ChatGPT demonstrated how generative AI could be used in education, business, healthcare, and software development.
2025	Agentic AI	AI systems evolved from answering questions to planning, reasoning, and performing multi-step tasks autonomously using tools, memory, and workflows.	AI agents can research topics, book appointments, analyze data, write reports, and coordinate multiple tasks with minimal human supervision.

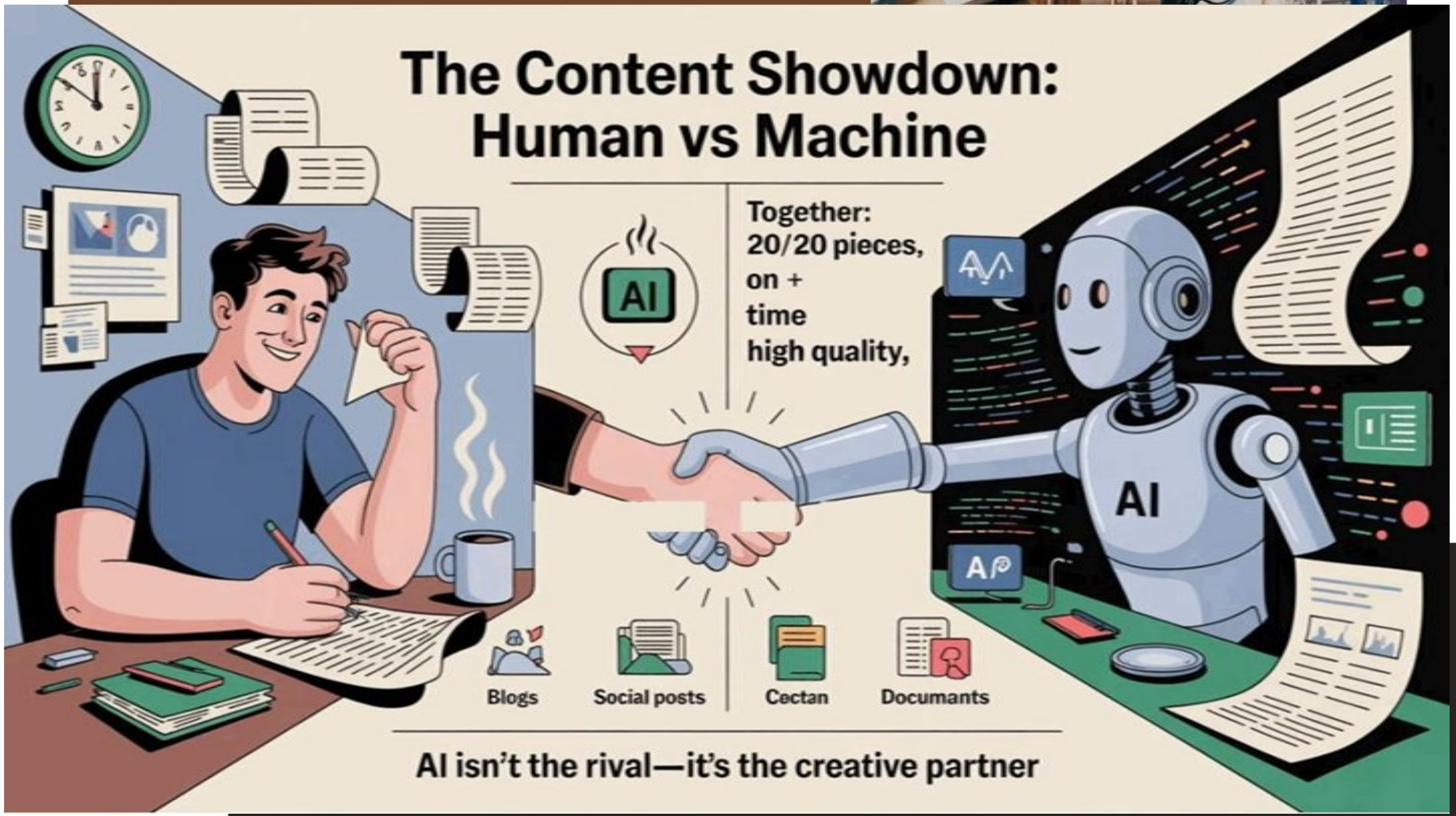






BERT | GPT




The Content Showdown:
Human vs Machine
AI
Together:
20/20 pieces,
on +
time
high quality,
AI
Blogs
Social posts
Cectan
Documants
AI isn't the rival—it's the creative partner

What is Artificial Intelligence?

- **Artificial Intelligence (AI) refers to computer systems capable of performing tasks that normally require human intelligence.**

- **Examples:**
  - Image recognition
  - Speech recognition
  - Language understanding
  - Decision making
  - Content generation



Comparative Chart: Capability-Based AI

Feature	ANI	AGI	ASI
Intelligence	Task-specific	Human-level	Beyond human
Learning	Limited	General	Unlimited
Self-awareness	No	Possible	Yes
Decision Making	Programmed	Human-like	Superior
Current Availability	Available	Under development	Theoretical
Examples	Alexa, ChatGPT	Human-like Robot	Hypothetical AI

PowerPoint Slide Show - [Day1] - PowerPoint

Comparative Chart: Functionality-Based AI

Not hearing anything? Turn up volume ×

Type	Memory	Learns	Self-aware	Current Status
Reactive Machine	No	No	No	Available
Limited Memory	Yes	Yes	No	Available
Theory of Mind	Yes	Yes	Partial	Research
Self-Aware	Yes	Yes	Yes	Theoretical

IBM Deep Blue chess compition

Recommendation system, Self driving cars

Understand emotion, Human intentions,social interaction

Comparison of Capability vs Functionality Classification

Capability Classification	Functionality Classification
Measures intelligence level	Measures working mechanism
ANI, AGI, ASI	Reactive, Limited Memory, Theory of Mind, Self-aware
Focuses on "How intelligent?"	Focuses on "How AI works?"
Based on capability	Based on behavior

Comparative Chart: Functionality-Based AI

Type	Memory	Learns	Self-aware	Current Status
Reactive Machine	No	No	No	Available
Limited Memory	Yes	Yes	No	Available
Theory of Mind	Yes	Yes	Partial	Research
Self-Aware	Yes	Yes	Yes	Theoretical

Comparison of Capability vs Functionality Classification

Capability Classification	Functionality Classification
Measures intelligence level	Measures working mechanism
ANI, AGI, ASI	Reactive, Limited Memory, Theory of Mind, Self-aware
Focuses on "How intelligent?"	Focuses on "How AI works?"
Based on capability	Based on behavior

Not hearing anything? Turn up volume ×

-Autonomy

Goals for AI:

-simulate human intelligence
-Automate repetative task
-Improvement on decision
-solve complex problems
-Understanding different languages
-REcognize patterns
-predict future events
-Assist human
-Enhance productivity
-Build intellegence autonomous systems

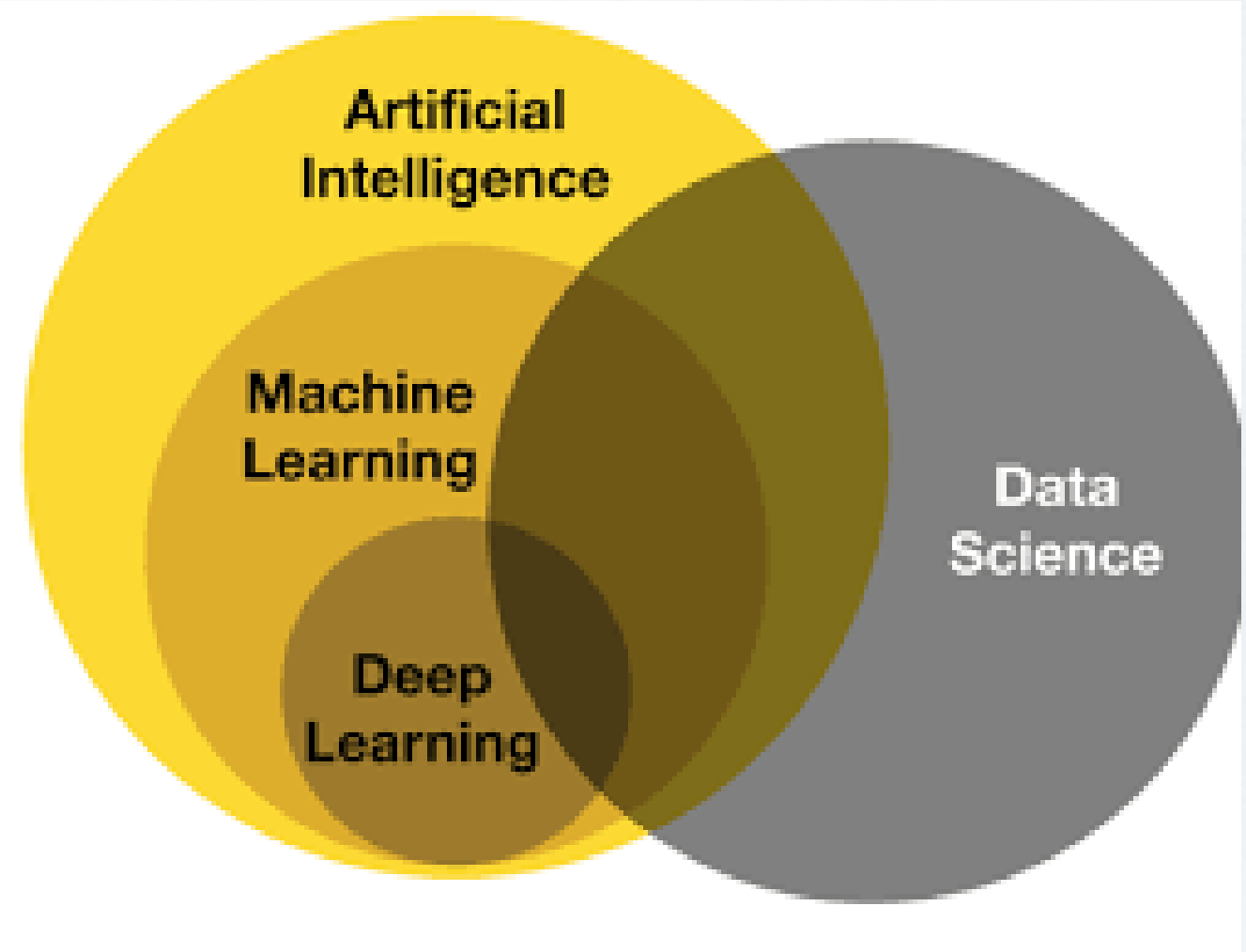
Without AI	AI
Manual work	Automation
Slow process	High acuracy
Human error	Faster decision
Limited stability	24 X 7 availability fo
Time consuming	Better predictions

4. Examples of Artificial Intelligence

AI System	Category	Application
ChatGPT	Generative AI	Conversation, writing, coding
Google Assistant	Voice AI	Personal assistant
Siri	Voice AI	Mobile assistant
Alexa	Voice AI	Smart home automation
Tesla Autopilot	Autonomous AI	Self-driving assistance
Netflix	Recommendation AI	Movie suggestions
Spotify	Recommendation AI	Music recommendations
Google Translate	NLP	Language translation
Face ID	Computer Vision	Biometric authentication
Grammarly	NLP	Grammar correction
Midjourney	Generative AI	AI image generation
GitHub Copilot	Coding AI	Programming assistance

Importance of Artificial Intelligence

Covers industry-relevant domains



- Machine Learning & Deep Learning
- Natural Language Processing (NLP)
- Computer Vision & Generative AI
- Data Analytics & Visualization

Importance of AI Hierarchy

Relationship Between the Technologies

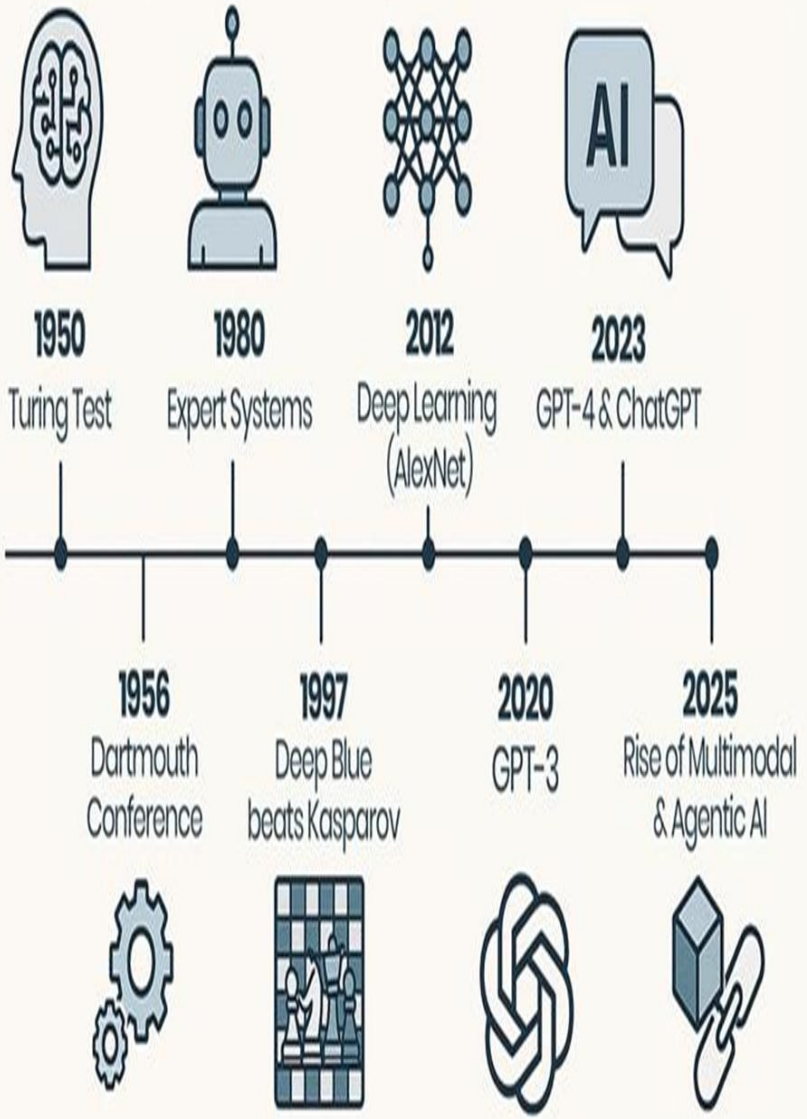
- **Artificial Intelligence (AI):** The overall field of making machines intelligent.
- **Machine Learning (ML):** A subset of AI where systems learn from data.
- **Deep Learning (DL):** A subset of ML that uses deep neural networks to solve complex problems.
- **Generative AI:** A subset of deep learning focused on creating original content rather than only analyzing or predicting.



Key Takeaway

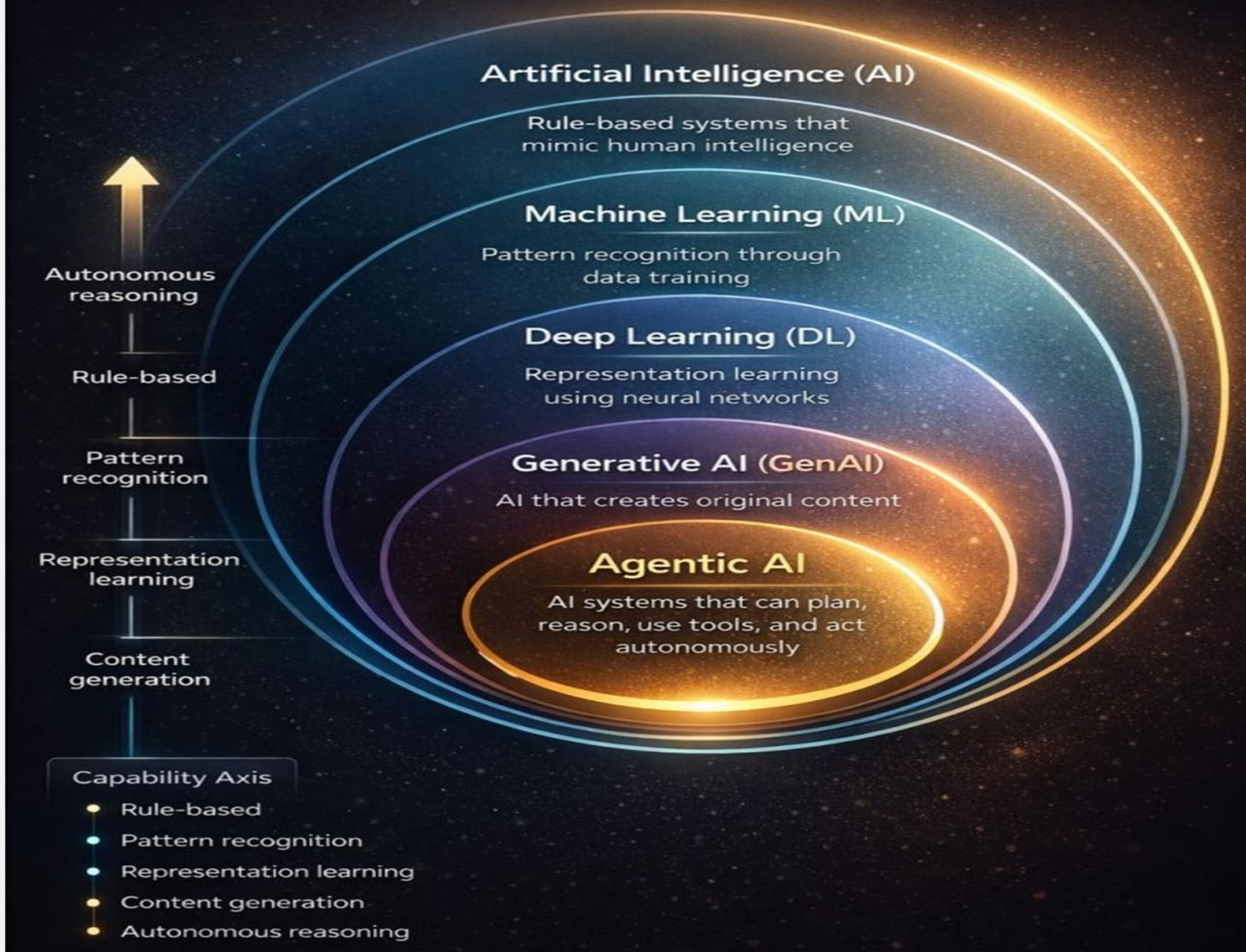
- AI enables machines to think and make decisions.
- ML allows machines to learn from experience (data).
- DL helps machines understand complex information such as images, speech, and natural language.
- Generative AI empowers machines to create new text, images, audio, video, and code, making it one of the most transformative AI technologies today.

HISTORY OF ARTIFICIAL INTELLIGENCE



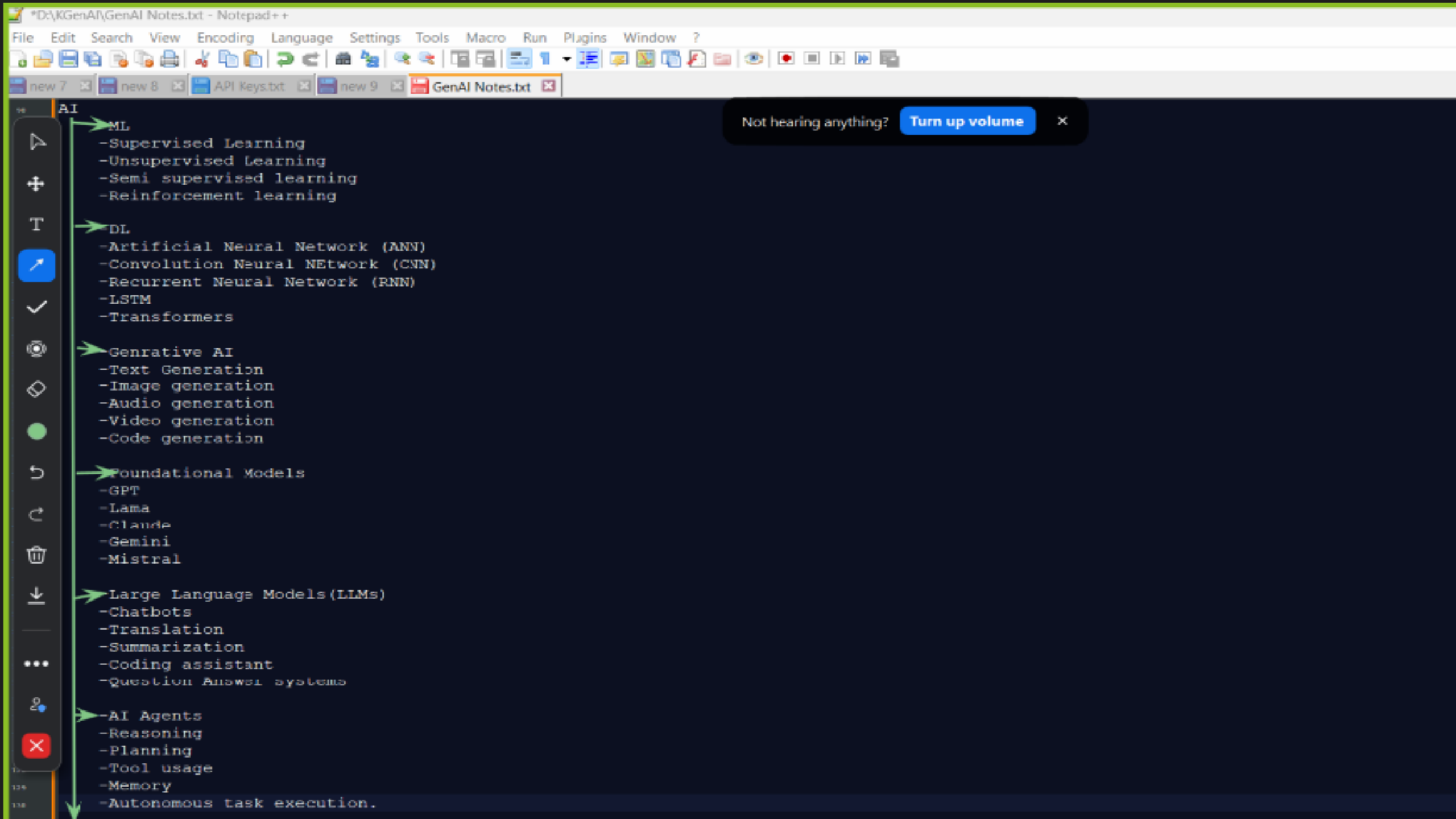
Evolution of AI Capabilities

From Intelligence to Autonomous Agents



Relationship Between the Layers

Layer	Purpose	Examples
Artificial Intelligence	Broad field of intelligent systems	Robotics, Expert Systems
Machine Learning	Learns patterns from data	Decision Trees, SVM
Deep Learning	Multi-layer neural networks	CNN, Transformer
Generative AI	Creates new content	ChatGPT, DALL·E
Foundation Models	Large pretrained models	GPT, Gemini, Llama
Large Language Models	Language-focused foundation models	GPT-4, Claude, Llama 3
AI Agents	Autonomous systems that reason, plan, and act	Coding agents, research agents, workflow automation


*D:\KGenAI\GenAI Notes.txt - Notepad++
File Edit Search View Encoding Language Settings Tools Macro Run Plugins Window ?
new 7
new 8
API Keys.txt
new 9
GenAI Notes.txt
Not hearing anything?
Turn up volume
AI
-ML
-Supervised Learning
-Unsupervised Learning
-Semi supervised learning
-Reinforcement learning
-DL
-Artificial Neural Network (ANN)
-Convolution Neural NEtwork (CNN)
-Recurrent Neural Network (RNN)
-LSTM
-Transformers
-Genrative AI
-Text Generation
-Image generation
-Audio generation
-Video generation
-Code generation
-Foundational Models
-GPT
-Lama
-Claude
-Gemini
-Mistral
-Large Language Models(LLMs)
-Chatbots
-Translation
-Summarization
-Coding assistant
-Question Answer systems
-AI Agents
-Reasoning
-Planning
-Tool usage
-Memory
-Autonomous task execution.